Coding & Solution

Date	7 July 2026
Team ID	SWUID2026221619
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Vinaymohan1234/A-Comprehensive-Measure-of-Well-Being
Programming Language(s)	Python 3.x
Framework(s) Used	Flask 3.x, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	1. HDI Score Prediction (Linear Regression, R-squared = 93.74%) 2. Country dropdown with Label Encoded values 3. 5-feature input form (Country, Life Exp, Schooling, GNI, Internet) 4. HDI Category classification (Very High / High / Medium / Low) 5. 3 Flask routes: home, prediction form, predict result 6. EDA: Strip plots, Correlation Heatmap, Actual vs Predicted plot 7. Pickle model serialization (HDI.pkl)
Pending / Incomplete Features	1. User authentication system 2. Batch CSV prediction upload 3. Historical country comparison charts 4. Database integration for prediction logging
Setup / Run Instructions	1. Install: pip install -r requirements.txt 2. Run notebook: jupyter notebook notebooks/hdi_model.ipynb (runs Epics 2–7) 3. Run Flask: python app.py 4. Open browser: http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Key Code Snippets

Epic	Code	Description
Epic 2	import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns	Import all required libraries
Epic 3	Development = pd.read_csv('HDI.csv') Development.head()	Load and inspect the dataset
Epic 4	X = Development.iloc[:,[2,5,6,7,67]] y = Development.iloc[:,4].values X = X.fillna(X.mean()) le = LabelEncoder() X['Country'] = le.fit_transform(X['Country'])	Feature selection, null handling, label encoding
Epic 5	x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)	90/10 train-test split
Epic 6	reg = LinearRegression() reg.fit(x_train, y_train) y_pred = reg.predict(x_test) r2_score(y_test, y_pred) # → 0.9374	Train model and evaluate
Epic 7	pickle.dump(reg, open('HDI.pkl','wb'))	Save model with Pickle
Epic 8	model = pickle.load(open('model/HDI.pkl','rb')) output = model.predict(df) y_pred = round(output[0][0],2)	Load model and predict in Flask